Sequences and Basic Series Convergence Tests

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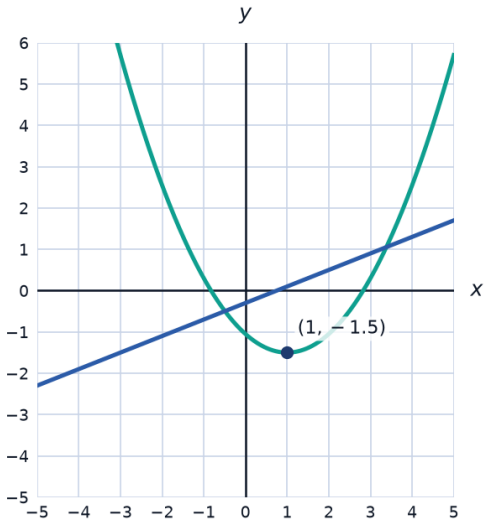
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the interval of convergence of $\sum_{n=0}^{\infty} x^n$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{2^n}$.

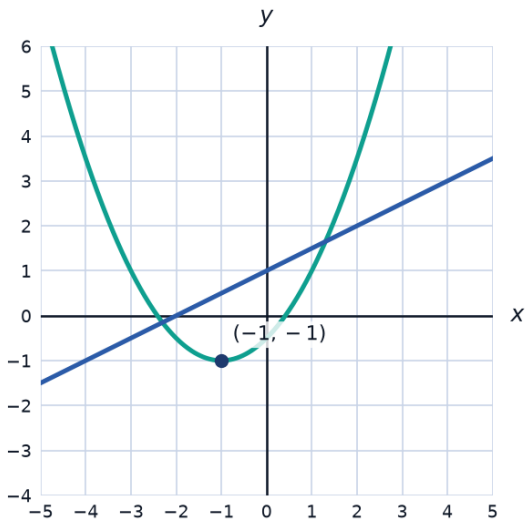


3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converge?

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the speed of a particle with $x = t^2,\ y = t^2$ at $t = 1$.



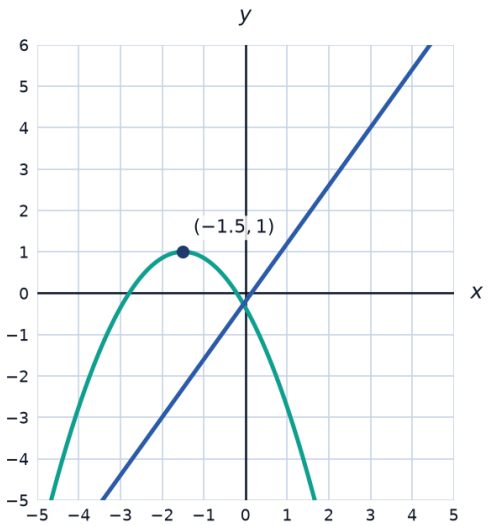
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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Does $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge or diverge?

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Unit 10 · Infinite Sequences & Series — Convergence Tests · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · Advanced / AP FRQOn an AP Calculus BC exam question asking whether a series $\sum_{n=1}^{\infty} a_n$ converges, a student writes: "Since $\lim_{n\to\infty} a_n = 0$, the series converges." Evaluate the mathematical validity of this statement, identify the logical fallacy, and provide a standard counterexample that disproves it.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsFind the radius of convergence R for the power series $\sum_{n=0}^{\infty} n!\,(2x-1)^n$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 2 · MediumThe graph of f' consists of a semicircle of radius 2 centered at the origin, lying above the x-axis. If $f(0) = 5$, find the value of $f(2)$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 1 · FoundationsOn a Rate In / Rate Out FRQ, if rain falls into a cistern at rate $R(t)$ and is pumped out at rate $P(t)$, when is the amount of water in the cistern at a relative maximum?



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 1 · FoundationsHow many total questions are on the AP Calculus AB exam, and what percentage of the composite score comes from Free-Response questions?

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[AP Calculus AB Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · MediumFind the third-degree Taylor polynomial $P_3(x)$ for $f(x) = \sqrt{x}$ centered at $c = 4$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · Advanced / AP FRQLet $P_4(x)$ be the fourth-degree Maclaurin polynomial for $\cos(x)$. Use the Lagrange Error Bound to prove that \(|\cos(0.5) - P_4(0.5)|

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

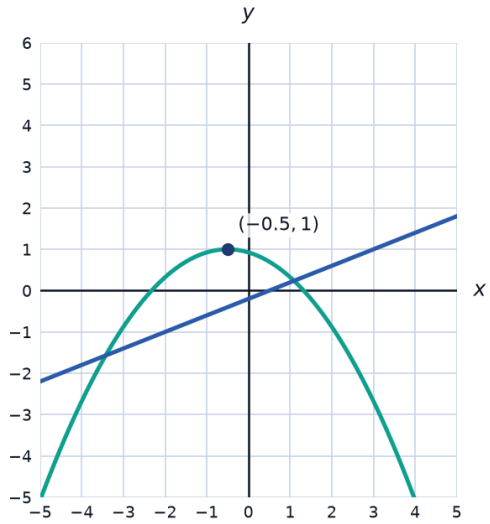
15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.02 Partial Fractions and Improper Integrals**

[Unit 10 · Power Series, Taylor & Maclaurin Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Euler's Method and Logistic Growth**

[AP Calculus BC Exam Format & Units Overview · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Parametric Equations**

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Vector-Valued Functions**

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Polar Curves and Derivatives**

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Area in Polar Coordinates**

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Determine the limit of the sequence $a_n = \frac{4n^2 - 3n}{2n^2 + 5}$.

a. 2

b. 4

c. ∞ (diverges)

d. 0

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which statement is guaranteed by the Monotone Convergence Theorem for Sequences?

a. Every bounded monotonic sequence converges

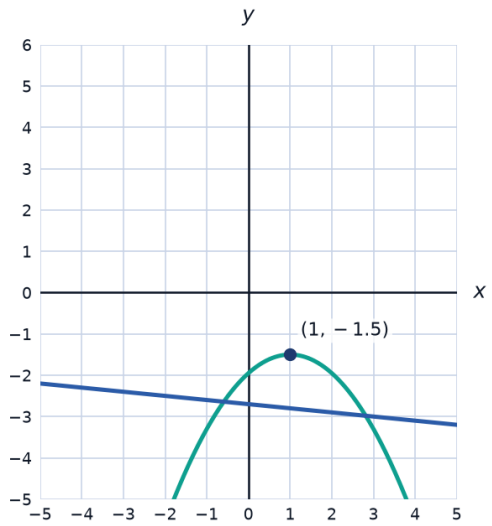
b. Every alternating sequence converges

c. Every bounded sequence is monotonic

d. Every convergent sequence is positive

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate the sum of the series $\sum_{n=1}^{\infty} 3(\frac{2}{5})^n$.



a. 2

b. $\frac{6}{5}$

c. $\frac{15}{2}$

d. $\frac{5}{2}$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\lim_{n\to\infty} a_n = \frac{1}{2}$, what does the n-th Term Test conclude about $\sum_{n=1}^{\infty} a_n$?

a. The series diverges

b. The series converges to $\frac{1}{2}$

c. The test is inconclusive

d. The series is geometric

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\lim_{n\to\infty} a_n = 0$, what can be concluded about $\sum a_n$?

a. The test is inconclusive; the series may converge or diverge

b. The series converges

c. The series diverges

d. The sum is zero

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$

b. $p \geq 1$

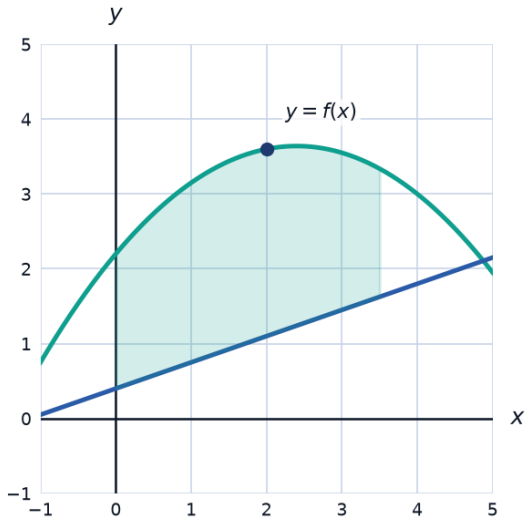
c. $p < 1$

d. $p > 0$

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which series converges by the Integral Test?



a. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

b. $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

c. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} \frac{1}{2n+1}$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What three conditions must $f(x)$ satisfy on $[1, \infty)$ to apply the Integral Test to $\sum a_n$?

a. $f(x)$ must be continuous, positive, and decreasing

b. $f(x)$ must be differentiable, alternating, and bounded

c. $f(x)$ must be positive and increasing

d. $f(x)$ must be a rational function

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5n}$.

a. Converges by Direct Comparison with $\sum \frac{1}{n^3}$

b. Diverges by Comparison with $\sum \frac{1}{n}$

c. Diverges by n-th term test

d. Inconclusive

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which series should be used for the Limit Comparison Test with $\sum_{n=1}^{\infty} \frac{3n+2}{2n^3-1}$?

a. $\sum \frac{1}{n^2}$ (convergent)

b. $\sum \frac{1}{n}$ (divergent)

c. $\sum \frac{1}{n^3}$ (convergent)

d. $\sum n$ (divergent)

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the Limit Comparison Test, if $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ where $0 < L < \infty$, what does this mean?

a. Both $\sum a_n$ and $\sum b_n$ either both converge or both diverge

b. $\sum a_n$ converges to L

c. $\sum a_n$ converges while $\sum b_n$ diverges

d. The test is inconclusive

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The alternating series $\sum (-1)^n \frac{1}{n}$:

a. Diverges

b. Converges absolutely

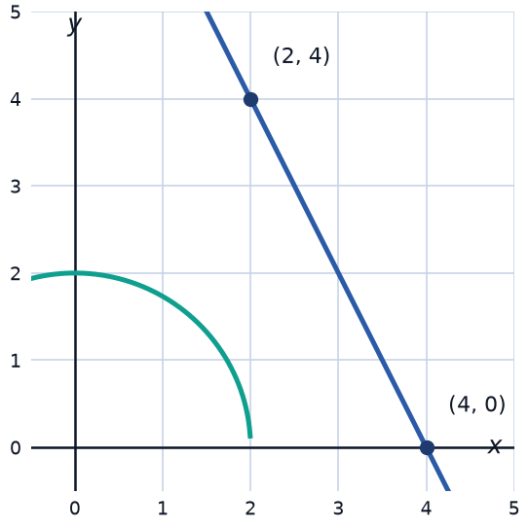
c. Converges conditionally

d. Oscillates without limit

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.



a. 4

b. 6

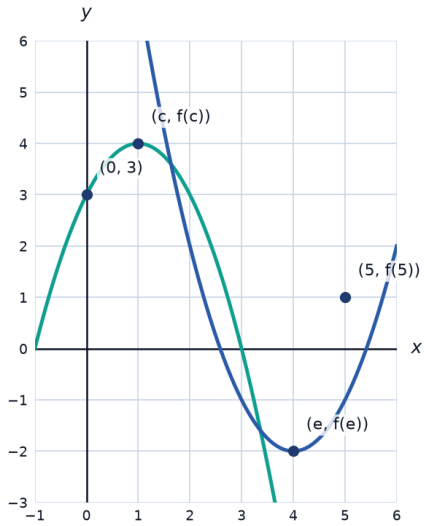
c. 5

d. 7

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. $If\int_{0}^{c} f(x) \, dx = k, whatisthevalueof\int_{c}^{5} f(x) \, dxintermsofk$?



a. 12 + k

b. 12 - k

c. k - 12

d. k + 2

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{(x-6)^n}{n \cdot 4^n}$.

a. ∞

b. $\frac{1}{4}$

c. 6

d. 4

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

What is the radius of convergence of the Maclaurin series for $\frac{1}{1-x}$? (variant 3)

a. ∞

b. 1

c. 2

d. 0

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A differentiable function f is defined on the interval $[-4, 4]$. The graph of f', the derivative of f, is shown in the figure above. The graph of f' consists of two line segments and a semicircle. (a) Find the value of $f(2) - f(-2)$. Show the setup for your answer. (b) For how many values of c in the interval $(-4, 4)$ does $f'(c) = 0$? Explain your reasoning. (c) The function f has a local maximum at $x = k$. Find the value of k. Explain your reasoning. (d) The function f has an inflection point at $x = p$. Find the value of p. Explain your reasoning.

a. There are no values of c for which $f'(c) = 0$.

b. There is one value of c for which $f'(c) = 0$.

c. There are two values of c for which $f'(c) = 0$.

d. There are three values of c for which $f'(c) = 0$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Vector-Valued Functions

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0?

a. $t = \frac{3}{2}$

b. $t = 1$

c. $t = 2$

d. $t = 3$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Polar Curves and Derivatives

Consider the function $f(x)$ whose graph is shown above. The graph of $f(x)$ consists of line segments and a quarter circle. Let $g(x) = \int_0^x f(t)\,dt$.

a. g'(2) = 2

b. g'(2) = 0

c. g'(5) = -2

d. g'(5) = 0

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Area in Polar Coordinates

If $f(x) = x^2 + 3$ is increasing on $[1, 6]$, a left Riemann sum with equal subintervals will:

a. equal the true area exactly

b. underestimate the true area

c. overestimate the true area

d. cannot be determined

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

If $f(x) = 2x^2 + 5$ is increasing on $[2, 5]$, a right Riemann sum with equal subintervals will:

a. equal the true area exactly

b. cannot be determined

c. underestimate the true area

d. overestimate the true area

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Using a right Riemann sum with 4 subintervals of equal width on $[0, 2]$, approximate $\int_0^2 (x^2 + x - 2)\,dx$.

a. $\frac{21}{4}$

b. $-\frac{3}{4}$

c. $-\frac{15}{4}$

d. $\frac{9}{4}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Using the Trapezoidal Rule with 2 subintervals on $[0, 6]$, approximate $\int_0^6 (2x^2 - 3x - 1)\,dx$.

a. 183

b. 21

c. 104

d. 102

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Integration by Parts

The table shows values of $g(x)$ at $x = [0, 3, 6, 8]$: $g = [2, 3, 8, 7]$. Using a right Riemann sum with these subintervals, approximate $\int_0^8 g(x)\,dx$.

a. 41

b. 47

c. 31

d. 50

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The table shows values of $g(x)$ at $x = [0, 2, 5, 7]$: $g = [7, 8, 12, 4]$. Using a left Riemann sum with these subintervals, approximate $\int_0^7 g(x)\,dx$.

a. 56

b. 65

c. 62

d. 60

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